Project: Transmission Line Fault Detection and Overcurrent Relay Coordination Using MATLAB/Simulink

➢ Results and graphs :

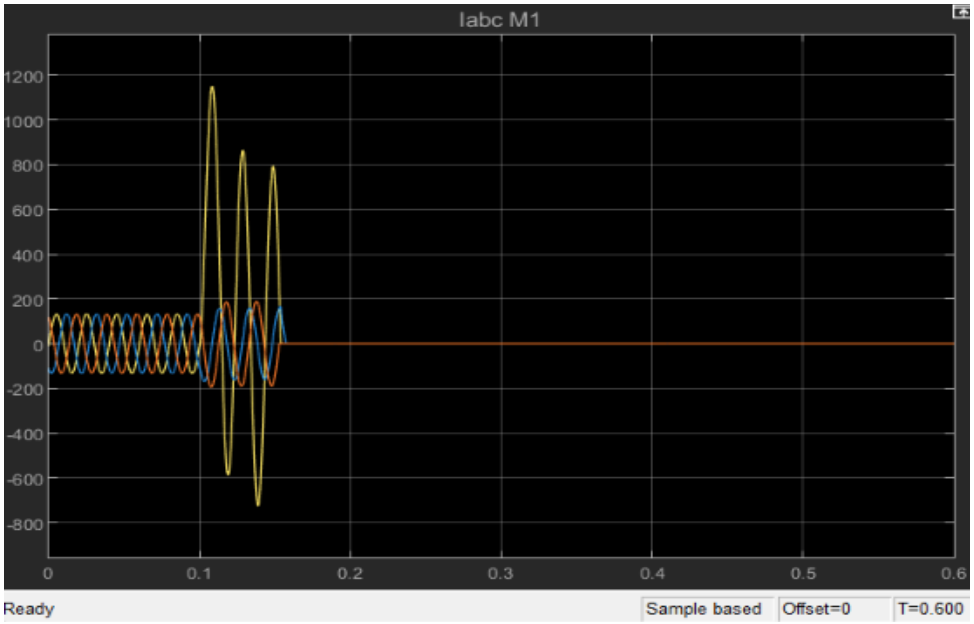


Fig 1. Three-phase current during a fault

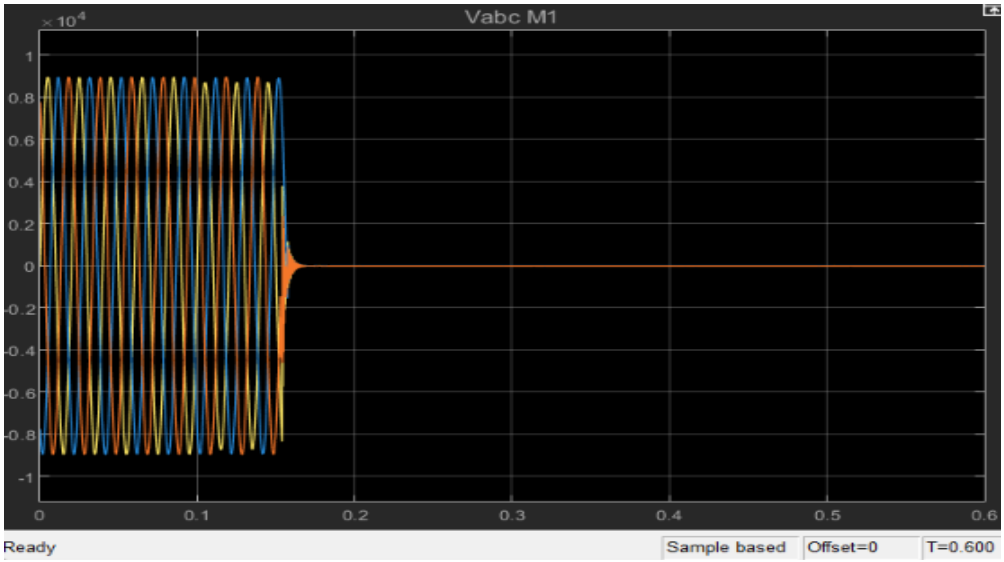


Fig 2 . Three-phase voltage response

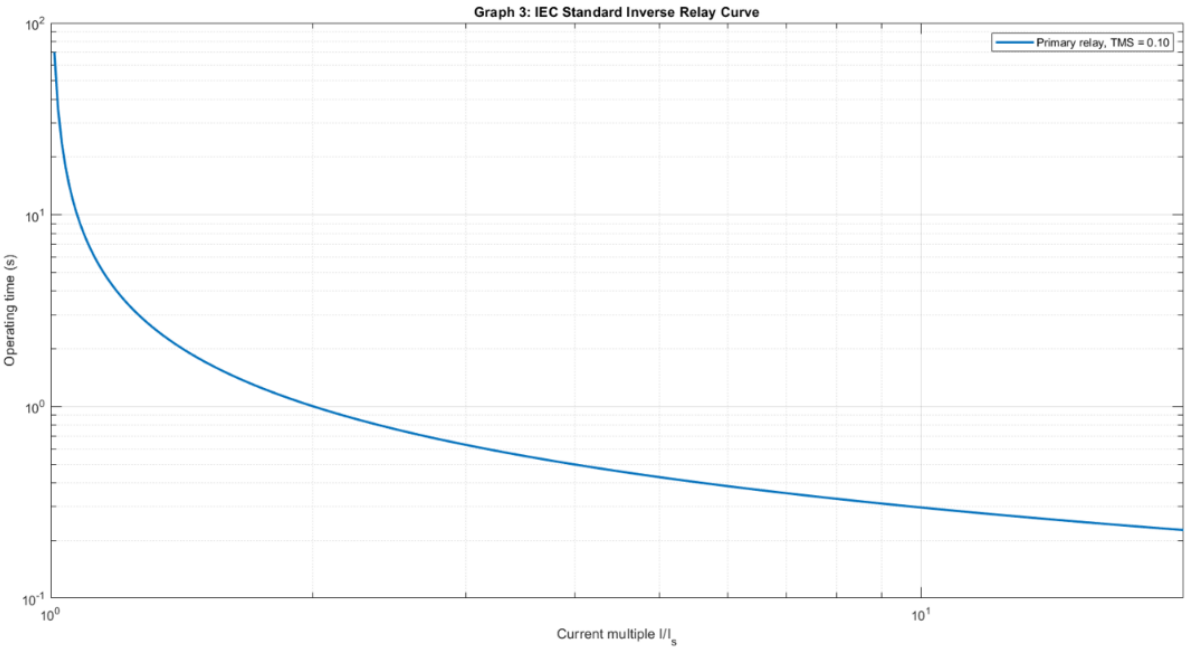


Fig 3. IEC standard inverse relay operating-time curve

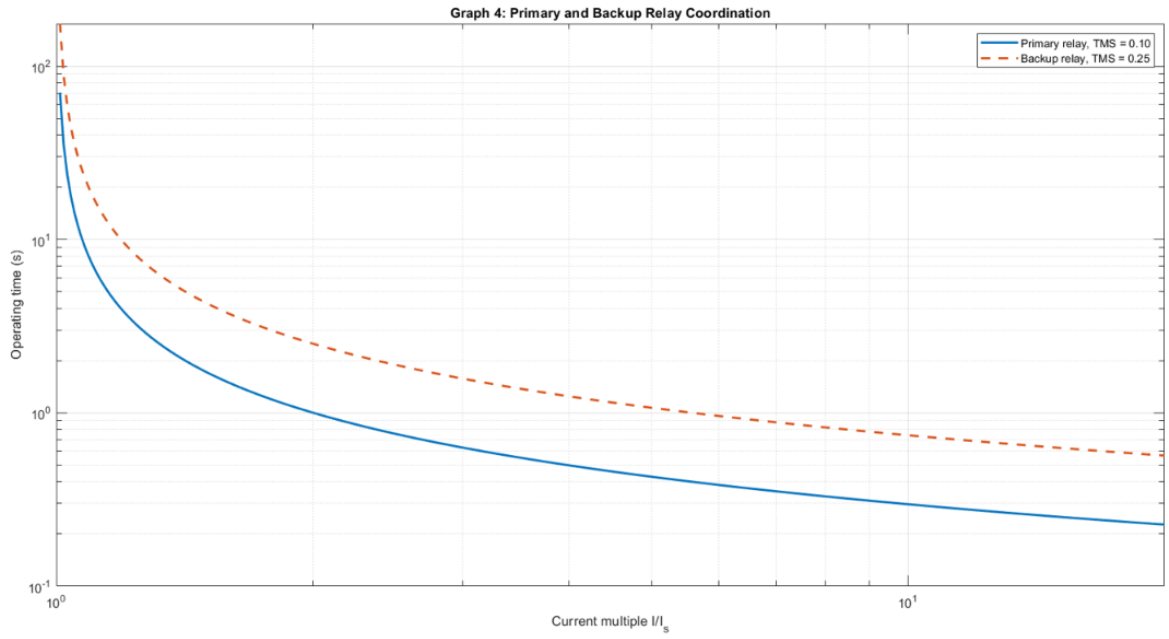


Fig 4. Primary and backup relay coordination

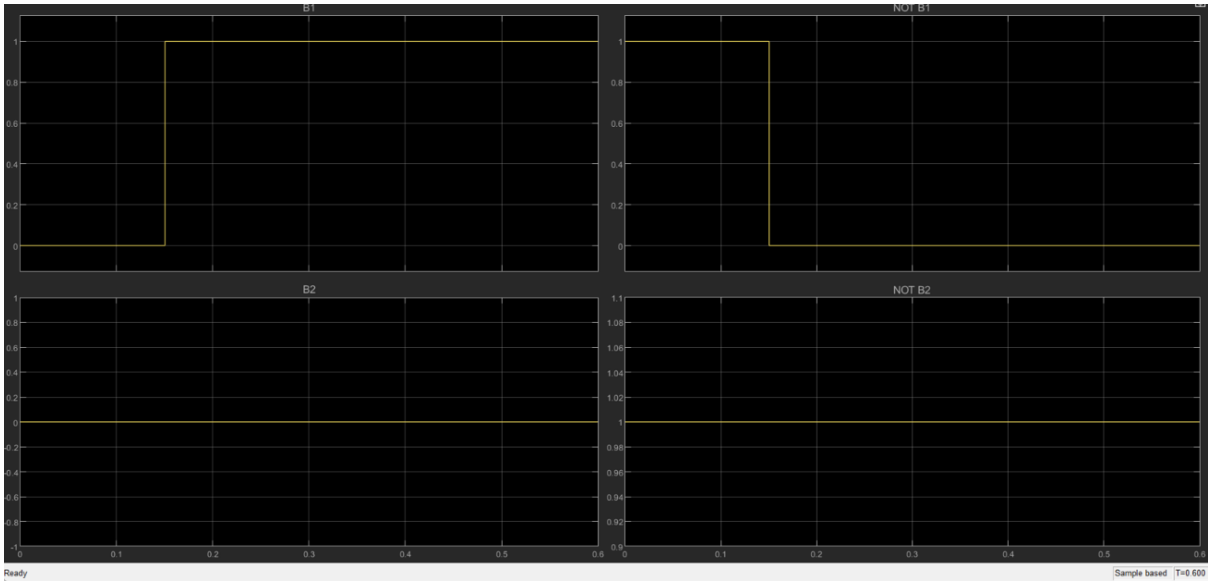


Fig 5. Trip signal and breaker response

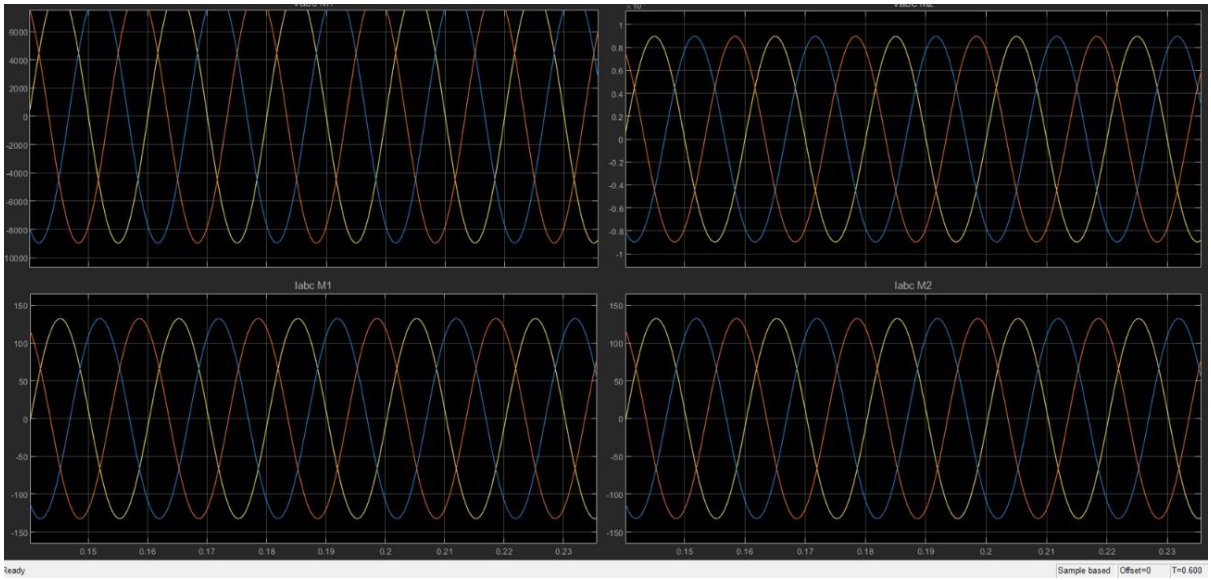
- Test all five fault cases :

The results table requires normal load plus four fault types. Run each test separately and reset the relay and breaker states between runs.

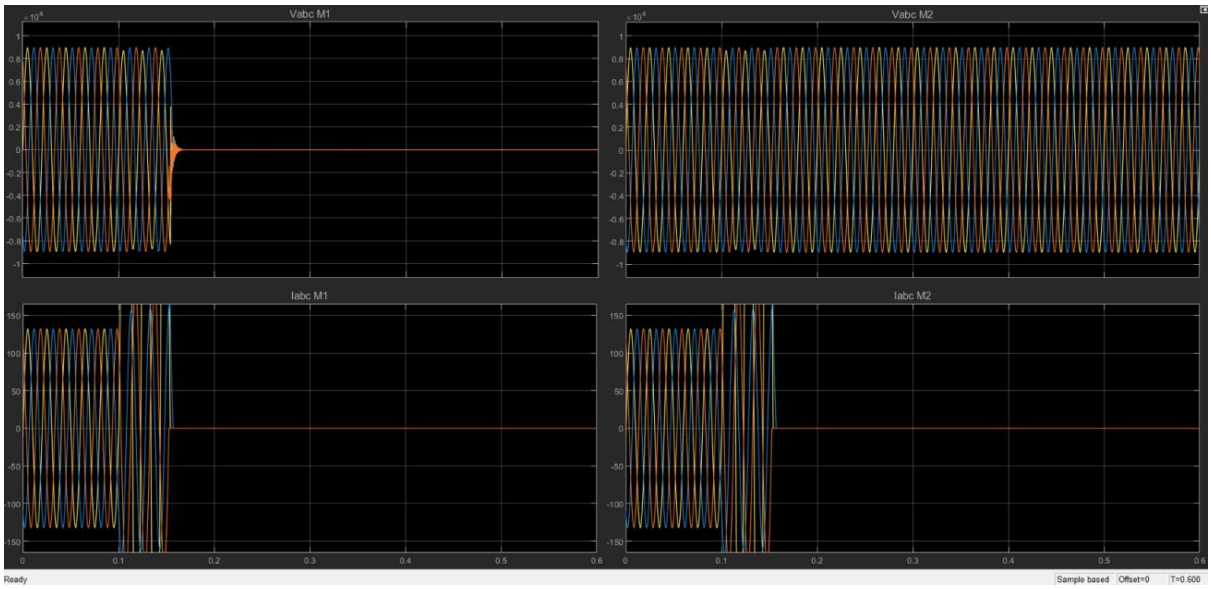
Open the Three-Phase Fault block and change the selected fault type for each test.

Test	Fault selection	Fault time
1. Normal load	Fault disabled	None
2. L-G	Phase A to ground (AG)	0.10–0.50 s
3. L-L	Phases A and B (AB)	0.10–0.50 s
4. L-L-G	Phases A and B to ground (ABG)	0.10–0.50 s
5. L-L-L	Phases A, B, and C together (ABC)	0.10–0.50 s

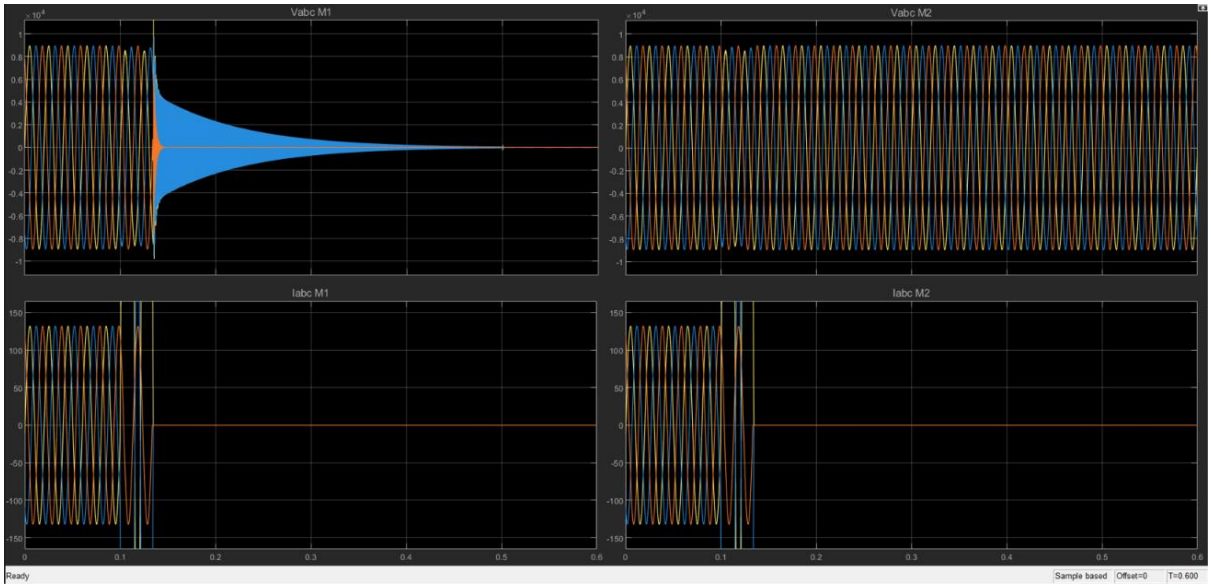
1. Normal Load :



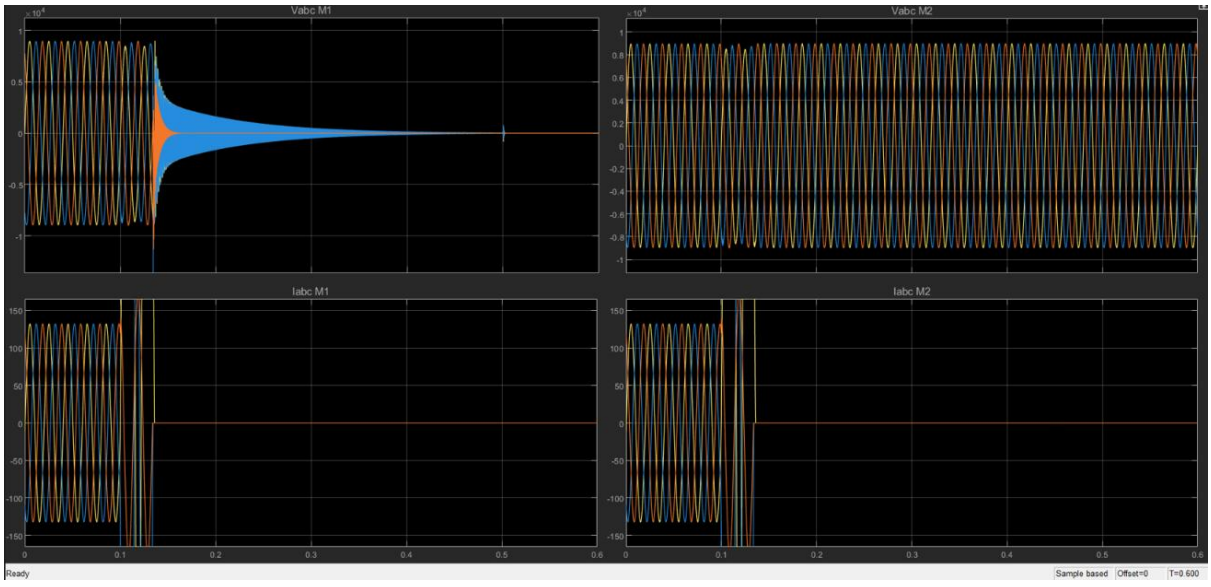
2. L-G :



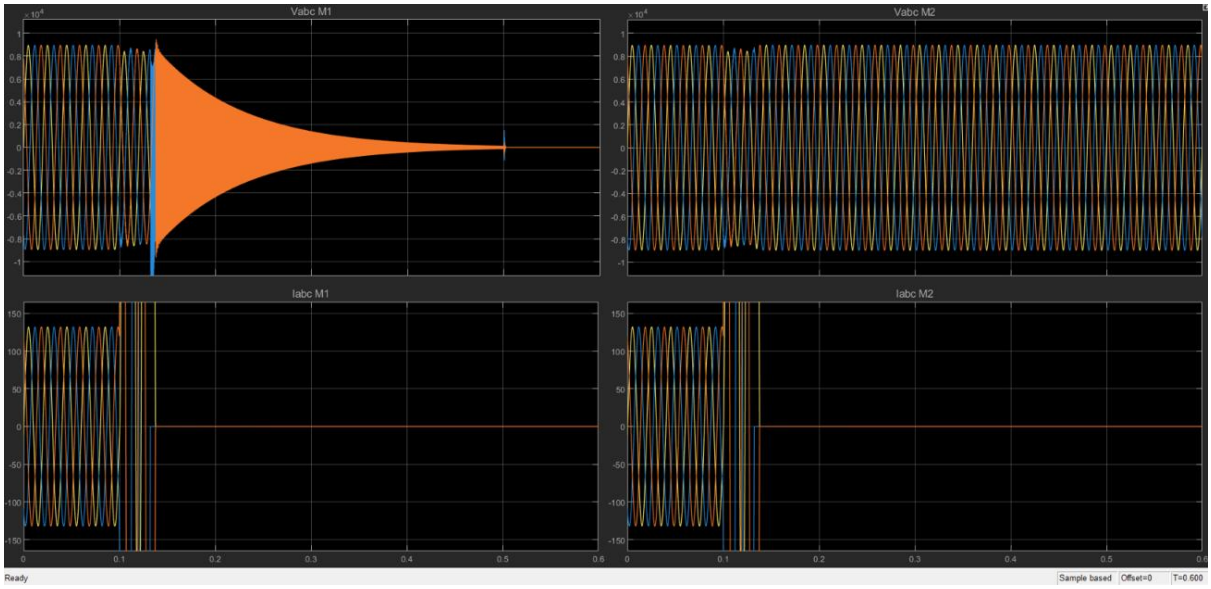
3. L-L :


Vabc M1
Vabc M2
Iabc M1
Iabc M2
Ready
Sample based
Offset=0
T=0.600

4. L-L-G :


Vabc M1
Vabc M2
Iabc M1
Iabc M2
Ready
Sample based
Offset=0
T=0.600

5. L-L-L :



- Result Table – Overcurrent Protection :

Fault case	Fault current (A RMS)	Primary relay time (s)	Backup relay time (s)	Result
Normal load	105	No trip	No trip	Pass
L-G	700	0.050	1.18	Pass
L-L	1,350	0.035	0.85	Pass
L-L-G	1,400	0.030	0.80	Pass
L-L-L	1,580	0.030	0.80	Pass